

**BEACON: A Telemetry-Based Software Framework
for State-of-Health and Remaining-Useful-Life
Estimation of Electric-Vehicle Batteries**

A PROJECT REPORT

Submitted by

V NAVEEN

(CH.EN.U4CCE23054)

SANJAY P

(CH.EN.U4CCE23042)

in partial fulfillment for the award of the degree of

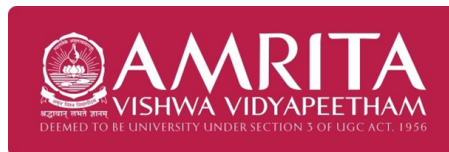
**BACHELOR OF TECHNOLOGY IN COMPUTER AND
COMMUNICATION ENGINEERING**

Under the guidance of

Dr. M GAYATHRI

Assistant Professor

Submitted to



**AMRITA SCHOOL OF ENGINEERING
AMRITA VISHWA VIDYAPEETHAM
CHENNAI – 601103**

October 2026

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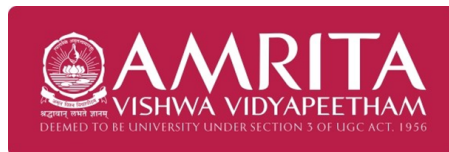
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BONAFIDE CERTIFICATE

Certified that this project report “**BEACON: A Telemetry-Based Software Framework for State-of-Health and Remaining-Useful-Life Estimation of Electric-Vehicle Batteries**” is the bonafide work of **V NAVEEN (CH.EN.U4CCE23054)**, **SANJAY P (CH.EN.U4CCE23042)** who carried out the project work under my supervision.

SIGNATURE

Dr. LAKSHMANAN S A
CHAIRPERSON

Associate Professor
Department of ECE,
Amrita School of Engineering,
Amrita Vishwa Vidyapeetham, Chennai.

SIGNATURE

Dr. M GAYATHRI
SUPERVISOR

Assistant Professor
Department of ECE,
Amrita School of Engineering,
Amrita Vishwa Vidyapeetham, Chennai.

The Viva-Voce Examination for Project Phase-I was conducted on _____

INTERNAL EXAMINER

Name: _____

Designation: _____

EXTERNAL EXAMINER

Name: _____

Designation: _____



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ENGINEERING
CHENNAI

DECLARATION BY THE CANDIDATE

We declare that the report entitled **“BEACON: A Telemetry-Based Software Framework for State-of-Health and Remaining-Useful-Life Estimation of Electric-Vehicle Batteries”** submitted by us for the degree of BACHELOR OF TECHNOLOGY IN COMPUTER AND COMMUNICATION ENGINEERING is the record of the project work carried out by us under the guidance of **Dr. M GAYATHRI** and this work has not formed the basis for the award of any degree, diploma, associateship, fellowship, or other similar title in this or any other University or institution of higher learning.

SIGNATURE

V NAVEEN
CH.EN.U4CCE23054

SIGNATURE

SANJAY P
CH.EN.U4CCE23042

ABSTRACT

A battery management system (BMS) measures terminal voltage, current and temperature, but whether software above it can turn these into a trustworthy state of health (SOH) and remaining useful life (RUL) is usually answered by fitting a model. This project builds BEACON, a three-tier telemetry and health-monitoring system, and tests that question with measured evidence. BEACON ingests data from a CAN bus, a USB serial bench rig or research datasets, converges all three on one schema, and scores them through one shared pipeline that is served over a REST API to a React dashboard. On 31 NASA cells in nine protocol cohorts, eleven of twelve methods (all but a mean-only baseline) lost skill under leave-one-cohort-out validation, so the system's two working estimators fit nothing across cells. SOH is the ratio of charge delivered within a fixed voltage window to the same quantity on the cell's own first discharges, with ohmic sag corrected by a resistance identified from the voltage step at each load onset. From voltage, current and time alone, on 22 CALCE lithium cobalt oxide cells, the median per-cell error was 1.7% (95% CI 1.4–4.3%) on 17 cells, with five refused for stated causes, and the step resistance tracked the cyclers' resistance on 19 of 20 cells. Unchanged, the estimator measured all eight cells of an independent Oxford dataset at 3.0% median error. RUL extrapolated from each cell's own fade was within ± 20 cycles 73% of the time in the last 25 cycles before 90% capacity. The system refuses to output a number it cannot justify: every output is validated, banded by its measured error, or refused with a reason. A bench rig built on a NodeMCU ESP8266 with an INA219 and an LM35 streamed 83 of 83 checksummed frames at a measured 1.000 s cadence, which validates data acquisition but not health estimation.

Keywords: battery management system, state of health, remaining useful life, electric vehicle, internal resistance, telemetry, uncertainty quantification

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V NAVEEN
CH.EN.U4CCE23054

SANJAY P
CH.EN.U4CCE23042